

# The rkdiagram Package

Reed-Kellogg Sentence Diagrams for  $\text{\LaTeX}$

Version 1.0

## Introduction

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The `rkdiagram` package typesets traditional Reed-Kellogg sentence diagrams in  $\text{\LaTeX}$ . Each word is tagged with a command describing its grammatical role; the package computes the layout and draws everything with `TikZ`. No natural language parsing or processing occurs. The user should have a working knowledge of parts of speech for best results.

**Features:** baselines, dividers, modifier diagonals, prepositional phrases, appositives, compound elements (including compound objects of prepositions), subordinate clauses, compound sentences, gerunds, and participles. Basic spacing and placement controls are available.

**Requirements:** Written using  $\text{\Lua\LaTeX}$  (but should work with any modern  $\text{\LaTeX}$  engine) plus `tikz`, `xparse`, `etoolbox`, and `pgfopts`.

## Loading the Package

```
\usepackage[
  spacing      = 1.3cm,
  modDepth     = 1.0cm,
  modgap       = 0.25cm,
  cmpSep       = 0pt,
  subClauseDepth = 2.5cm,
  linewidth    = 0.5pt,
  font         = \rmfamily,
  color        = black,
]{rkdiagram}
```

The optional values above are the defaults. Override any option mid-document with `\rkset`:

```
\rkset{spacing=1.5cm, color=blue}
```

## Package options

| Option                      | Default    | Effect                                                                                              |
|-----------------------------|------------|-----------------------------------------------------------------------------------------------------|
| <code>spacing</code>        | 1.2cm      | Horizontal gap (padding) between baseline elements                                                  |
| <code>modDepth</code>       | 1.0cm      | Fixed horizontal step between plain adj/adv modifiers; minimum slant length for all modifiers/preps |
| <code>modgap</code>         | 0.25cm     | Extra vertical headspace between nested modifier levels                                             |
| <code>cmpSep</code>         | 0pt        | Minimum half-gap between compound platforms (0pt = auto)                                            |
| <code>subClauseDepth</code> | 2.5cm      | Vertical drop to a subordinate clause baseline                                                      |
| <code>gerundPost</code>     | 1.2cm      | Minimum post height for gerund T-platforms                                                          |
| <code>linewidth</code>      | 0.4pt      | Thickness of all diagram lines                                                                      |
| <code>font</code>           | (doc font) | Font applied to all diagram words                                                                   |
| <code>color</code>          | black      | Line and text colour                                                                                |
| <code>bridgeY</code>        | 0pt        | Pin compound-sentence bridge y (0pt = auto)                                                         |

## The rkdiagram Environment

Every diagram lives inside `rkdiagram`, which opens a *TikZ* picture:

```
\begin{rkdiagram}[spacing=1.5cm] % optional per-diagram options
  \rkSubject{word}
  \rkVerb{word}
\end{rkdiagram}
```

Words are automatically centred on their baseline lines. The diagram sits on the text baseline and can appear inline or in its own paragraph.

## Baseline Elements

Baseline elements sit on the main horizontal line, placed left to right in the order written.

### Subject and Verb

A full vertical bar separates subject from verb. The bar's ascender (above the baseline) is full height; its descender (below the baseline) is half that height.

*“Dogs bark.”*

```
\begin{rkdiagram}
  \rkSubject{Dogs}
  \rkVerb{bark}
\end{rkdiagram}
```

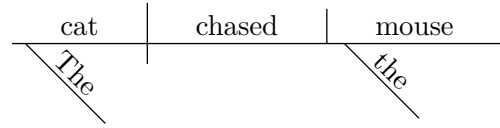


## Direct Object

`\rkDObj` places a short bar (above baseline only) after the verb.

*“The cat chased the mouse.”*

```
\begin{rkdiagram}
  \rkSubject[\rkAdj{The}]{cat}
  \rkVerb{chased}
  \rkDObj[\rkAdj{the}]{mouse}
\end{rkdiagram}
```

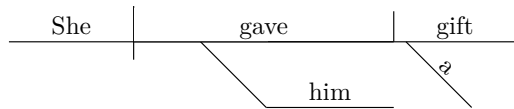


## Indirect Object

`\rkIObj` draws a diagonal platform below the verb midpoint.

*“She gave him a gift.”*

```
\begin{rkdiagram}
  \rkSubject{She}
  \rkVerb{gave}
  \rkIObj{him}
  \rkDObj[\rkAdj{a}]{gift}
\end{rkdiagram}
```

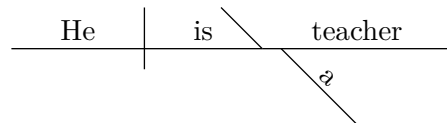


## Predicate Nominative and Predicate Adjective

After a linking verb, a back-diagonal divider (`\`) introduces a predicate nominative (`\rkPredNom`) or predicate adjective (`\rkPredAdj`). The diagonal slants at 45°, matching the modifier diagonals, and its bottom touches the baseline without descending below it.

*“He is a teacher.”*

```
\begin{rkdiagram}
  \rkSubject{He}
  \rkVerb{is}
  \rkPredNom[\rkAdj{a}]{teacher}
\end{rkdiagram}
```



## Modifiers

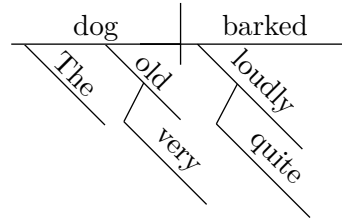
Modifiers hang below the word they modify on a 45° diagonal. Use `\rkAdj` for adjectives and articles, `\rkAdv` for adverbs. Pass them as the *optional first argument* of the word they modify, comma-separated:

```
\rkSubject[
  \rkAdj{The},
  \rkAdj{old}
]{dog}
```

A modifier takes its own optional argument for sub-modifiers, creating chains of diagonals:

*“The very old dog barked quite loudly.”*

```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkAdj[\rkAdv{very}]{old}
  ]{dog}
  \rkVerb[
    \rkAdv[\rkAdv{quite}]{loudly}
  ]{barked}
\end{rkdiagram}
```



The `modgap` option adds extra vertical clearance between nested levels (default 0.25 cm), preventing crowding when modifier chains are deep.

**Modifier spacing:** plain adjectives and adverbs (no sub-modifiers) advance the sibling cursor by exactly `modDepth`, giving even horizontal spacing regardless of label width. Their slant lines are still sized to fit the word — long labels simply extend past the slant ends, matching hand-drawn convention. Modifiers that carry sub-modifiers (e.g. `\rkAdj[\rkAdv{very}]{old}`) use the word-fit formula so the slant is long enough to keep the sub-modifier connector clear of the next sibling. Use `modDepth` to control the global step between plain modifiers.

## Prepositional Phrases

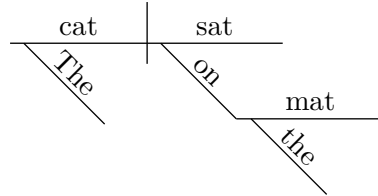
`\rkPrep` draws a diagonal with the preposition label, then a horizontal line for the object.

```
\rkPrep[mods-on-object]{preposition}{object}
```

Place it in the modifier list of the word it modifies:

*“The cat sat on the mat.”*

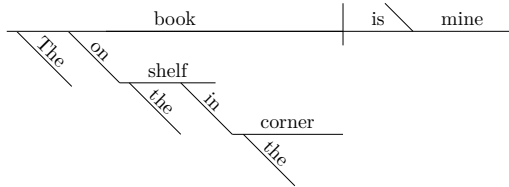
```
\begin{rkdiagram}
  \rkSubject[\rkAdj{The}]{cat}
  \rkVerb[
    \rkPrep[\rkAdj{the}]{on}{mat}
  ]{sat}
\end{rkdiagram}
```



Prepositional phrases nest freely — a prep modifying the object of another prep:

*“The book on the shelf in the corner is mine.”*

```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkPrep[
      \rkAdj{the},
      \rkPrep[\rkAdj{the}]{in}{corner}
    ]{on}{shelf}
  ]{book}
  \rkVerb{is}
  \rkPredNom{mine}
\end{rkdiagram}
```



## Compound Structures

### Basic usage

`\rkCompound` joins two words or phrases with a conjunction. The full signature is:

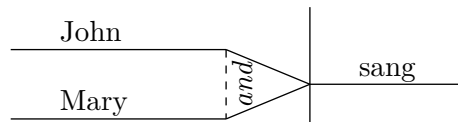
```
\rkCompound[conj][mods_a]{word_a}[mods_b]{word_b}
```

- `[conj]` — conjunction label (default: *and*)
- `[mods_a]` / `[mods_b]` — optional per-word modifier lists (same syntax as baseline commands)
- `{word_a}` / `{word_b}` — the two words or phrases

Each word sits on its own horizontal platform. The platforms are connected by convergence slants meeting at a point, a dashed vertical bar at the junction, and the conjunction written between the bar and the convergence point. Vertical separation scales with the conjunction width automatically.

*“John and Mary sang.”*

```
\begin{rkdiagram}
  \rkSubject{\rkCompound[and]{John}{Mary}}
  \rkVerb{sang}
\end{rkdiagram}
```

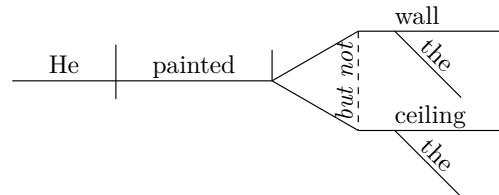


### Per-word modifiers

Modifiers on each compound word go in the optional modifier-list arguments:

*“He painted the wall but not the ceiling.”*

```
\begin{rkdiagram}
  \rkSubject{He}
  \rkVerb{painted}
  \rkDObj{
    \rkCompound[but not]
      [\rkAdj{the}]{wall}
      [\rkAdj{the}]{ceiling}
  }
\end{rkdiagram}
```



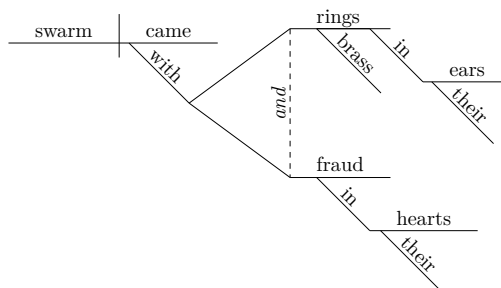
Modifiers placed in the list (e.g. `\rkAdj`, `\rkAdv`, `\rkDObj`) draw as proper diagonals on each word’s platform. Slot commands like `\rkDObj` inside a per-word modifier list attach a direct object to that compound word’s platform line.

## Compound object of a preposition

Pass `\rkCompound` as the object argument of `\rkPrep` to fork the preposition into two objects:

“...with brass rings in their ears and fraud in their hearts...”

```
\rkPrep{with}{  
  \rkCompound[and]  
  [  
    \rkAdj{brass},  
    \rkPrep[\rkAdj{their}]{in}{ears}  
  ]  
  {rings}  
  [\rkPrep[\rkAdj{their}]{in}{hearts}]  
  {fraud}  
}
```



Use `cmpSep` when modifiers on the compound words are deep and need extra vertical room between platforms.

## Compound Sentences

Use `\rkCompoundSent` *standalone* (not inside `rkdiagram`) to diagram two independent clauses in a single picture, connected by a dashed-line bridge with the coordinating conjunction. Each clause body must be wrapped in its own `rkdiagram` environment:

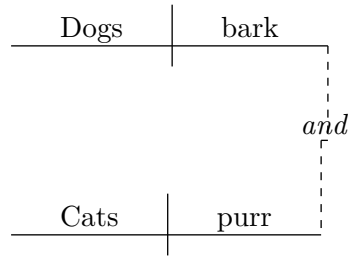
```
\rkCompoundSent[conj][sep][opts1]  
  {\begin{rkdiagram}body1\end{rkdiagram}}  
  [opts2]  
  {\begin{rkdiagram}body2\end{rkdiagram}}
```

- `[conj]` — conjunction (default: *and*)
- `[sep]` — vertical distance between the two baselines (default: 3 cm)
- `[opts1]` / `[opts2]` — optional per-clause `\rkset`-style options
- `{body1}` / `{body2}` — full `rkdiagram` environments for each clause

The connector drops a dashed vertical from each clause’s verb to a midpoint bridge, draws a short horizontal line there, and labels it with the conjunction.

*“Dogs bark and cats purr.”*

```
\rkCompoundSent[and][2.5cm]
{\begin{rkdiagram}
  \rkSubject{Dogs}
  \rkVerb{bark}
\end{rkdiagram}}
{\begin{rkdiagram}
  \rkSubject{Cats}
  \rkVerb{purr}
\end{rkdiagram}}
```

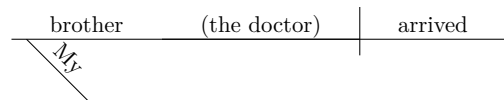


## Appositives

`\rkAppos` places an appositive in parentheses on the baseline, directly to the right of the word it renames. Modifier commands inside `\rkAppos` render inline.

*“My brother, the doctor, arrived.”*

```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{My},
    \rkAppos{\rkAdj{the} doctor}
  ]{brother}
  \rkVerb{arrived}
\end{rkdiagram}
```



## Subordinate Clauses

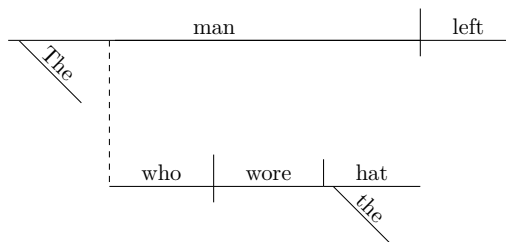
`\rkSubClause` draws a complete sub-baseline below the main one, connected by a dashed vertical. It appears in the modifier list of the word it modifies. The optional argument is the relative or subordinating word placed first on the sub-baseline; it doubles as the grammatical subject, so do *not* add a separate `\rkSubject` call for it.

```
\rkSubClause[rel-word]{body}
```



“The man who wore the hat left.”

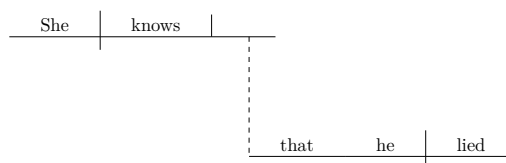
```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkSubClause[who]{
      \rkVerb{wore}
      \rkDObj[\rkAdj{the}]{hat}
    }
  ]{man}
  \rkVerb{left}
\end{rkdiagram}
```



A noun clause as direct object — the sub-clause body starts directly with `\rkVerb`:

“She knows that he lied.”

```
\begin{rkdiagram}
  \rkSubject{She}
  \rkVerb{knows}
  \rkDObj{
    \rkSubClause[that]{
      \rkSubject{he}
      \rkVerb{lied}
    }
  }
\end{rkdiagram}
```



## Gerunds

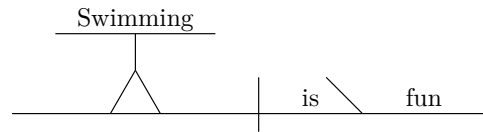
A *gerund* is a verb form used as a noun. It sits on a stepped T-platform: a full-width horizontal foot at the noun-slot baseline, two short angled legs rising to an apex, a vertical post, and a short word-platform at the top. Use `\rkGerund` as the *word argument* of any baseline command (`\rkSubject`, `\rkDObj`, `\rkPredNom`, etc.) or as the object of `\rkPrep`.

```
\rkGerund[mods]{word}
```

The optional `[mods]` argument holds modifiers of the gerund itself (adjectives, prepositional phrases, etc.). The package pre-scans the list to compute a post height tall enough to keep all modifier slants clear of the foot. Use the `gerundPost` option (default 1.2cm) to set a minimum post height globally.

*“Swimming is fun.”*

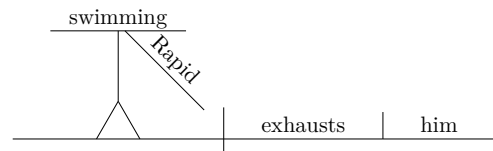
```
\begin{rkdiagram}
  \rkSubject{\rkGerund{Swimming}}
  \rkVerb{is}
  \rkPredAdj{fun}
\end{rkdiagram}
```



Gerund with its own modifiers — the post height adjusts automatically:

*“Rapid swimming exhausts him.”*

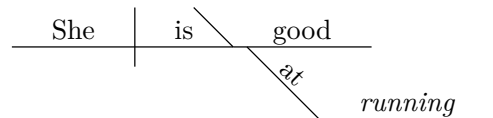
```
\begin{rkdiagram}
  \rkSubject{
    \rkGerund[\rkAdj{Rapid}]{swimming}
  }
  \rkVerb{exhausts}
  \rkDObj{him}
\end{rkdiagram}
```



Gerund as the object of a preposition:

*“She is good at running quickly.”*

```
\begin{rkdiagram}
  \rkSubject{She}
  \rkVerb{is}
  \rkPredAdj[
    \rkPrep{at}{
      \rkGerund[\rkAdv{quickly}]{running}
    }
  ]{good}
\end{rkdiagram}
```



## Participles

A *participle* functions as an adjective modifying a noun. It is drawn with a hockey-stick shape: a 45° diagonal (word label above it), a smooth curve at the elbow, and a horizontal stub. Use `\rkParticiple` inside a modifier list — the same position as `\rkAdj`.

```
\rkParticiple[tail]{word}
```

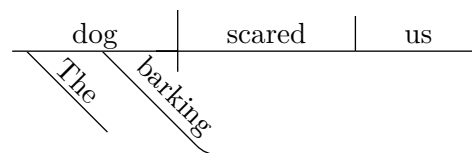
Without a tail argument the shape is simply the angled line with word label — enough for a bare participial adjective. With a tail, the horizontal stub serves as a mini-baseline for a participial direct object and/or modifiers of the participle:

```
% Participial object on the tail:
\rkPartObj[mods]{word}

% Adverb or prep phrase modifying the participle, also on the tail:
\rkAdv{quickly}
\rkPrep[obj-mods]{prep}{obj}
```

*“The barking dog scared us.”*

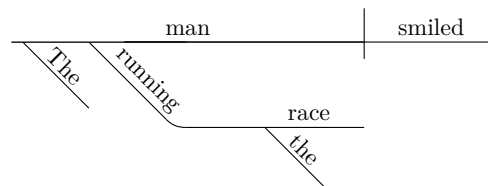
```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkParticiple{barking}
  ]{dog}
  \rkVerb{scared}
  \rkDObj{us}
\end{rkdiagram}
```



Participle with a participial object (`\rkPartObj`) on the tail:

*“The man running the race smiled.”*

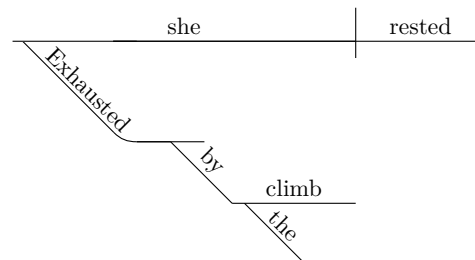
```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkParticiple[
      \rkPartObj[\rkAdj{the}]{race}
    ]{running}
  ]{man}
  \rkVerb{smiled}
\end{rkdiagram}
```



Participle with a prepositional phrase on the tail (the traditional passive-participial form):

*“Exhausted by the climb, she rested.”*

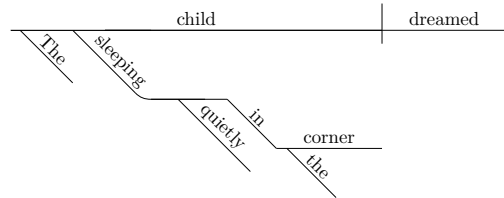
```
\begin{rkdiagram}
  \rkSubject[
    \rkParticiple[
      \rkPrep[\rkAdj{the}]{by}{climb}
    ]{Exhausted}
  ]{she}
  \rkVerb{rested}
\end{rkdiagram}
```



Multiple tail items (adverb + prepositional phrase) both branch from the horizontal stub:

*“The child sleeping quietly in the corner dreamed.”*

```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkParticiple[
      \rkAdv{quietly},
      \rkPrep[\rkAdj{the}]{in}{corner}
    ]{sleeping}
  ]{child}
  \rkVerb{dreamed}
\end{rkdiagram}
```



## Infinitives

An infinitive (*to + verb*) has two distinct uses:

- **Noun function** — used as subject, direct object, or predicate nominative, just like a gerund. The diagram uses the same stepped T-pedestal as `\rkGerund`, but the word *to* is written on the vertical post. Use `\rkInfinitive` wherever you would use `\rkGerund`: as the word argument of `\rkSubject`, `\rkDObj`, etc.
- **Modifier function** — used as adjective or adverb. The diagram looks like a prepositional phrase: a diagonal line with *to* written on it, and the verb on a short horizontal at the bottom. That horizontal is a full mini-baseline: use `\rkInfObj` to place a direct object, and `\rkAdv/\rkPrep` for adverbial modifiers. Use `\rkInfinitive` inside any modifier list.

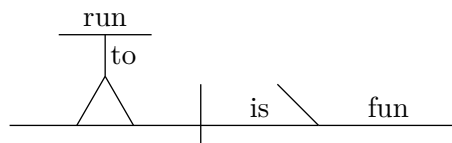
```
\rkInfinitive[arg]{verb}
```

The optional `arg` carries modifiers of the verb in the noun form; in the modifier form it is tail content (objects, adverbial modifiers) on the mini-baseline.

### Noun form — infinitive as subject or object

*“To run is fun.”*

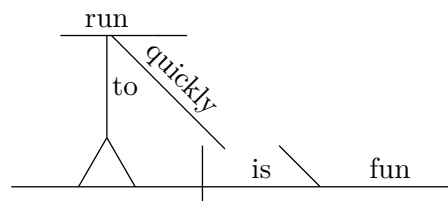
```
\begin{rkdiagram}
  \rkSubject{\rkInfinitive{run}}
  \rkVerb{is}
  \rkPredNom{fun}
\end{rkdiagram}
```



Infinitive noun with a modifier on the verb:

*“To run quickly is fun.”*

```
\begin{rkdiagram}
  \rkSubject{
    \rkInfinitive[\rkAdv{quickly}]{run}
  }
  \rkVerb{is}
  \rkPredNom{fun}
\end{rkdiagram}
```

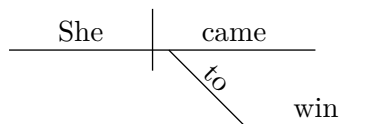


### Modifier form — infinitive as adjective or adverb

Without a tail the shape is a diagonal labeled *to* with the verb on a short horizontal — identical in form to a prepositional phrase.

*“She came to win.”*

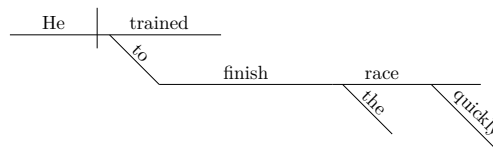
```
\begin{rkdiagram}
  \rkSubject{She}
  \rkVerb[\rkInfinitive{win}]{came}
\end{rkdiagram}
```



With tail content, use `\rkInfObj` for a direct object and `\rkAdv/\rkPrep` for adverbial modifiers of the infinitive verb:

*“He trained to finish the race quickly.”*

```
\begin{rkdiagram}
  \rkSubject{He}
  \rkVerb[
    \rkInfinitive[
      \rkInfObj[\rkAdj{the}]{race}
      \rkAdv{quickly}
    ]{finish}
  ]{trained}
\end{rkdiagram}
```



### Nominative Absolute

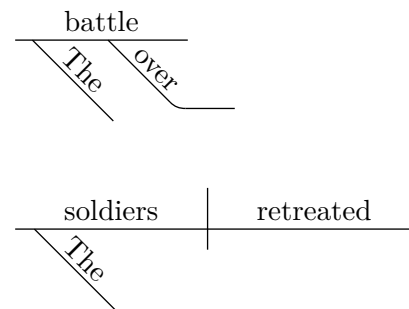
A nominative absolute is a noun-plus-participial-phrase that modifies the whole sentence rather than any single word. In a Reed-Kellogg diagram it floats above the main clause with no connector line.

```
\rkNomAbs[noun-mods]{noun}{ptcl-tail}{ptcl-word}
```

- **Call order:** `\rkNomAbs` can be called anywhere inside the diagram environment — it always floats above the main baseline at a fixed height.
- **Noun modifiers:** the optional `[noun-mods]` argument takes the same modifier syntax as `\rkSubject`’s modifier list (`\rkAdj`, etc.).
- **Participial tail:** the `{ptcl-tail}` argument is passed directly to `\rkParticiple` — use `\rkPartObj`, `\rkAdv`, or `\rkPrep` there; pass `{}` for an empty tail.
- **No participle:** leave `{ptcl-word}` empty for an absolute with an implied *being* (e.g. *his work done*).

“*The battle over, the soldiers retreated.*”

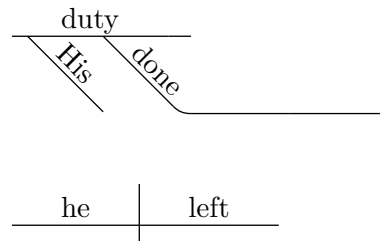
```
\begin{rkdiagram}
  \rkSubject[\rkAdj{The}]{soldiers}
  \rkVerb{retreated}
  \rkNomAbs[\rkAdj{The}]{battle}{}{over}
\end{rkdiagram}
```



Nominative absolute with a participial object on the tail:

“*His duty done, he left.*”

```
\begin{rkdiagram}
  \rkSubject{he}
  \rkVerb{left}
  \rkNomAbs[\rkAdj{His}]{duty}{
    \rkPartObj{
      }{done}
    }
\end{rkdiagram}
```



## Manual Spacing Controls

Three one-shot commands let you override automatic spacing for a single element; they reset to zero automatically after use.

```
\rkSlotWidth{len}  % min line width for the NEXT baseline slot
\rkSkip{len}       % gap in baseline BEFORE the next slot
\rkSlantLen{len}   % min diagonal length for the NEXT modifier/prep
```

### `\rkSlotWidth` — minimum word line length

Forces the next slot’s baseline line to be at least `len` wide, regardless of word width. Useful for aligning parallel diagrams or leaving annotation space.

*“Dogs bark. (4 cm subject line)”*

```
\begin{rkdiagram}
  \rkSlotWidth{4cm}
  \rkSubject{Dogs}
  \rkVerb{bark}
\end{rkdiagram}
```



### `\rkSkip` — gap before a slot

Inserts a gap of `len` in the baseline *before* the next slot. The gap is drawn as a continuation of the current baseline line, effectively pushing everything right from that point.

*“Dogs bark. (1.5 cm gap before verb)”*

```
\begin{rkdiagram}
  \rkSubject{Dogs}
  \rkSkip{1.5cm}
  \rkVerb{bark}
\end{rkdiagram}
```

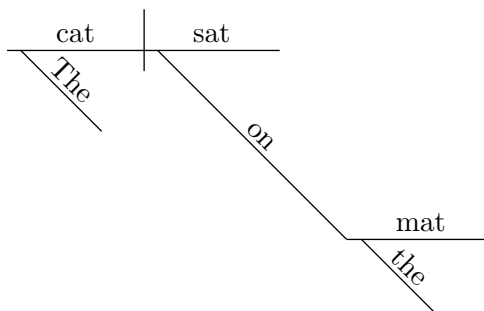


### `\rkSlantLen` — minimum diagonal length

Forces the next modifier or prepositional slant to be at least `len` long. Place it on the line immediately before the `\rkAdj`, `\rkAdv`, or `\rkPrep` call.

*“The cat sat on the mat. (longer prep slant)”*

```
\begin{rkdiagram}
  \rkSubject[\rkAdj{The}]{cat}
  \rkVerb[
    \rkSlantLen{2.5cm}
    \rkPrep[\rkAdj{the}]{on}{mat}
  ]{sat}
\end{rkdiagram}
```



All three commands work inside sub-clauses and on compound-word platforms as well as the main baseline.

### `\rkNomAbsHeight` — nominative absolute height

By default `\rkNomAbs` floats the absolute phrase at `subClauseDepth` above the main baseline. When the participle and its own modifiers are deep, they can crowd the main clause. Call `\rkNomAbsHeight` immediately before `\rkNomAbs` to override the height for that one phrase; it resets automatically.

```
\rkNomAbsHeight{4cm}
```

## Structural Depth Controls

The global options `modDepth`, `subClauseDepth`, and `cmpSep` control the vertical geometry of the diagram. Any of them can be changed per-diagram by passing options to the `rkdiagram` environment, or mid-document with `\rkset`:

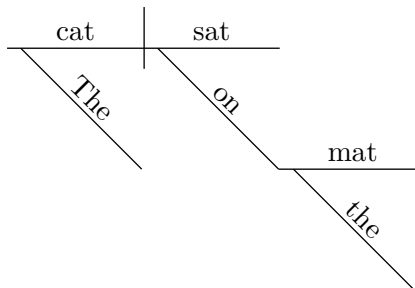
```
\begin{rkdiagram}[modDepth=1.5cm, subClauseDepth=3cm]
...
\end{rkdiagram}
```

### `modDepth` — modifier spacing and slant depth

Controls two related things simultaneously. For plain adjectives and adverbs (no sub-modifiers), it is the *fixed horizontal step* between siblings — all such modifiers are evenly spaced regardless of label width, with their slant lines sized to fit the word. For modifiers with sub-modifiers, it is the *minimum* slant length. Increase it to widen spacing and push labels further from the baseline; decrease it to tighten compact diagrams.

“*Deeper slants (`modDepth=1.6cm`)*”

```
\begin{rkdiagram}[modDepth=1.6cm]
\rkSubject[\rkAdj{The}]{cat}
\rkVerb[\rkPrep[\rkAdj{the}]{on}{mat}]{sat}
\end{rkdiagram}
```



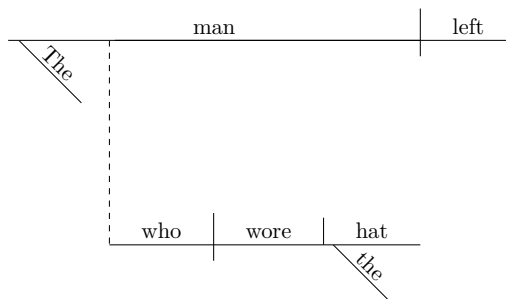
### `subClauseDepth` — vertical drop to sub-baseline

Controls how far below the main baseline a subordinate clause is drawn. Increase it when deep modifier chains on the main clause would overlap the sub-clause.



“Deeper sub-clause (*subClauseDepth=3.5cm*)”

```
\begin{rkdiagram}[subClauseDepth=3.5cm]
  \rkSubject[
    \rkAdj{The},
    \rkSubClause[who]{
      \rkVerb{wore}
      \rkDObj[\rkAdj{the}]{hat}
    }
  ]{man}
  \rkVerb{left}
\end{rkdiagram}
```

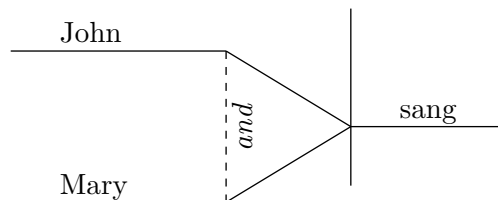


### cmpSep — compound platform separation

Sets the minimum vertical half-gap between the two platforms of a compound element. The default `Opt` lets the package scale the gap automatically from the conjunction width. Set an explicit value when modifier chains on the compound words need extra vertical clearance.

“Extra separation (*cmpSep=1.0cm*)”

```
\begin{rkdiagram}[cmpSep=1.0cm]
  \rkSubject{
    \rkCompound[and]{John}{Mary}
  }
  \rkVerb{sang}
\end{rkdiagram}
```



### bridgeY — connector bridge height (compound sentences)

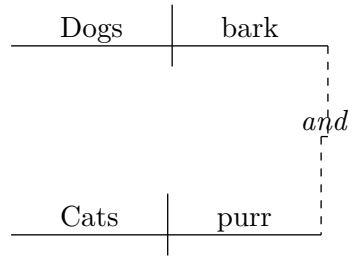
By default `\rkCompoundSent` places the horizontal bridge halfway between the two baselines, automatically lowering it if clause 1 modifiers extend into that region. Use `bridgeY` (a negative length, measured from clause 1’s baseline) to pin the bridge at an exact position:

```
\rkCompoundSent[and][3cm][bridgeY=-1.8cm]
{
  \begin{rkdiagram}
    \rkSubject{Dogs}
    \rkVerb{bark}
  \end{rkdiagram}
}
{
  \begin{rkdiagram}
    \rkSubject{Cats}
    \rkVerb{purr}
  \end{rkdiagram}
}
```

Pass `bridgeY` in the first per-clause options argument. Setting it to `Opt` restores automatic placement.

*“Bridge pinned at  $-1.2$  cm”*

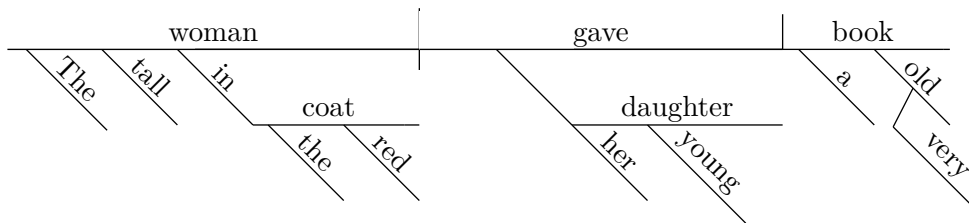
```
\rkCompoundSent[and][2.5cm]
[bridgeY=-1.2cm]
{\begin{rkdiagram}
\rkSubject{Dogs} \rkVerb{bark}
\end{rkdiagram}}
{\begin{rkdiagram}
\rkSubject{Cats} \rkVerb{purr}
\end{rkdiagram}}
```



## Complete Example

*“The tall woman in the red coat gave her young daughter a very old book.”*

```
\begin{rkdiagram}
\rkSubject[
  \rkAdj{The},
  \rkAdj{tall},
  \rkPrep[
    \rkAdj{the},
    \rkAdj{red}
  ]{in}{coat}
]{woman}
\rkVerb{gave}
\rkIObj[
  \rkAdj{her},
  \rkAdj{young}
]{daughter}
\rkDObj[
  \rkAdj{a},
  \rkAdj[\rkAdv{very}]{old}
]{book}
\end{rkdiagram}
```



## Command Reference

## Baseline commands

All baseline commands share the signature `\rkXxx[modifier-list]{word(s)}`. The modifier list is a comma-separated sequence of modifier, prep, appositive, or sub-clause calls.

| Command                             | Role                                             |
|-------------------------------------|--------------------------------------------------|
| <code>\rkSubject[mods]{word}</code> | Subject; left of the full vertical divider       |
| <code>\rkVerb[mods]{word}</code>    | Verb; right of the full vertical divider         |
| <code>\rkDObj[mods]{word}</code>    | Direct object; right of the short vertical bar   |
| <code>\rkIObj[mods]{word}</code>    | Indirect object; diagonal platform below verb    |
| <code>\rkPredNom[mods]{word}</code> | Predicate nominative; right of the back-diagonal |
| <code>\rkPredAdj[mods]{word}</code> | Predicate adjective; right of the back-diagonal  |

### Modifier and structural commands

| Command                                                       | Role                                                                                                                                                                        |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\rkAdj[mods]{word}</code>                               | Adjective/article; diagonal below modified noun                                                                                                                             |
| <code>\rkAdv[mods]{word}</code>                               | Adverb; diagonal below modified verb/adj/adv                                                                                                                                |
| <code>\rkPrep[obj-mods]{prep}{obj}</code>                     | Prep phrase; diagonal + horizontal object line                                                                                                                              |
| <code>\rkAppos{content}</code>                                | Appositive; parenthesized on the baseline                                                                                                                                   |
| <code>\rkSubClause[rel]{body}</code>                          | Subordinate clause; dashed connector to sub-baseline                                                                                                                        |
| <code>\rkCompound[conj][mA]{wA}[mB]{wB}</code>                | Compound element; forked platforms with conjunction                                                                                                                         |
| <code>\rkGerund[mods]{word}</code>                            | Gerund; stepped T-platform (used as word argument of any baseline command or <code>\rkPrep</code> object)                                                                   |
| <code>\rkInfinitive[arg]{verb}</code>                         | Infinitive; noun form (pedestal + <i>to</i> on post) when used as word argument; modifier form (diagonal labeled <i>to</i> + verb on horizontal) when used in modifier list |
| <code>\rkInfObj[mods]{word}</code>                            | Direct object on the infinitive mini-baseline; only inside <code>\rkInfinitive</code> 's tail argument                                                                      |
| <code>\rkParticiple[tail]{word}</code>                        | Participle; hockey-stick modifier of a noun (used inside modifier lists like <code>\rkAdj</code> )                                                                          |
| <code>\rkPartObj[mods]{word}</code>                           | Participial object on the tail; only inside <code>\rkParticiple</code> 's tail argument                                                                                     |
| <code>\rkNomAbs[noun-mods]{noun}{ptcl-tail}{ptcl-word}</code> | Nominative absolute; floating phrase above the main clause, no connector                                                                                                    |

### Compound sentence

| Command                                                             | Role                                                                                                           |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>\rkCompoundSent[conj][sep][opts1]{body1}[opts2]{body2}</code> | Compound sentence (standalone; each body wraps its own <code>\rkdiagram</code> ), connected by a dashed bridge |

## Manual spacing (one-shot)

| Command                           | Effect                                                         |
|-----------------------------------|----------------------------------------------------------------|
| <code>\rkSlotWidth{len}</code>    | Minimum line width for the next baseline slot                  |
| <code>\rkSkip{len}</code>         | Gap in baseline before the next slot                           |
| <code>\rkSlantLen{len}</code>     | Minimum diagonal length for the next modifier/prep             |
| <code>\rkNomAbsHeight{len}</code> | Height above main baseline for the next <code>\rkNomAbs</code> |

## Structural depth options (per-diagram or global)

These are standard package options (see §2) but can also be passed to any `rkdiagram` environment or to `\rkset` mid-document:

| Option                          | Effect                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------|
| <code>modDepth=len</code>       | Fixed step between plain adj/adv siblings; minimum slant length for sub-modified modifiers and preps |
| <code>subClauseDepth=len</code> | Vertical drop from main baseline to sub-clause baseline                                              |
| <code>cmpSep=len</code>         | Minimum half-gap between compound platforms (0pt = auto)                                             |
| <code>bridgeY=neg-len</code>    | Pin compound-sentence bridge at a fixed y below clause 1 (0pt = auto)                                |

## Global configuration

| Command                          | Effect                                   |
|----------------------------------|------------------------------------------|
| <code>\rkset{key=val,...}</code> | Override any package option mid-document |

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